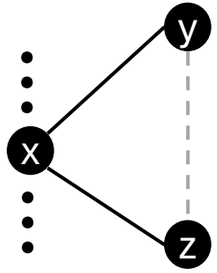
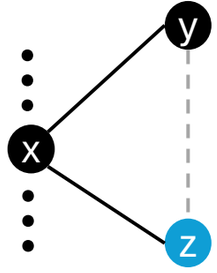
We assume valid wedges to have paths where |parent| < |child|; for all the cases below, |x| < |y|; local edge list is in increasing order, imputing |y| < |z| => |x| < |z|



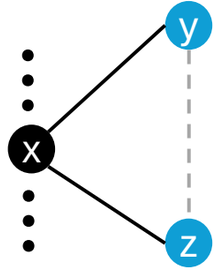
Case 1

PE(x) = PE(y) = PE(z)
Neighborhoods of y and z are local to x's PE; *no* communication



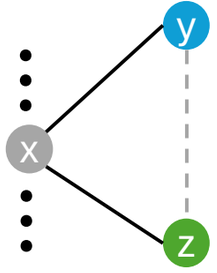
Case 2

PE(x) = PE(y) != PE(z)
Neighborhood of y is local to x's PE; *no* communication



Case 3

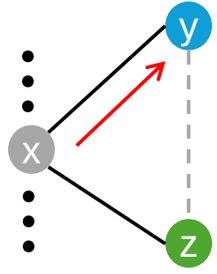
PE(x) != PE(y) = PE(z)
Neighborhoods of y and z in same remote PE; *likely* communication



Case 4

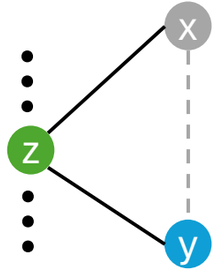
PE(x) != PE(y) != PE(z)
Neighbors of y and z in remote PEs; *likely* communication

Assume x-y-z is a triangle, and PE(x) < PE(y) < PE(z); how to ensure *single* counting?



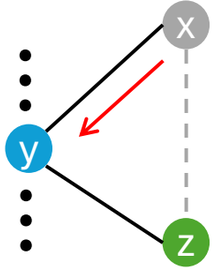
PE(x)'s POV

1. |x| < |y|, valid path
2. PE(x) sends message to PE(y) inquiring y-z



PE(z)'s POV

1. |z| ≮ |x|, invalid path



PE(y)'s POV

1. |y| ≮ |x|, invalid path
2. Receive from PE(x)
3. +1 count: y-z exists